



COURSE CODE/TITLE: CPE 212

SECTION: CPE31S2

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ACTIVITY NO. ____ Midterm Skills Exam

1. PROCEDURE OUTPUT

Task 1: Prepare the Server

```
mdmlot@workstation: ~/CPE212_loterte/midterm_skills
GNU nano 6.2 inventory.yaml
l:
children:
  ubuntu_servers:
    hosts:
      ubuntu-node:
        ansible_host: 192.168.56.110
  centos_servers:
    hosts:
      centos-node:
        ansible_host: 192.168.56.114
```

Task 2: Create the Prometheus User

```
# tasks file for prometheus_ubuntu
- name: create Prometheus system user
  ansible.builtin.user:
    name: prometheus
    shell: /bin/false
    system: true
    create_home: false
    state: present
    become: true

- name: create Prometheus group
  ansible.builtin.group:
    name: prometheus
    state: present
```

```
TASK [prometheus_ubuntu : create Prometheus system user] *****
ok: [ubuntu-node]
```

```
TASK [prometheus_ubuntu : create Prometheus group] *****
ok: [ubuntu-node]
```

```
TASK [prometheus_centos : create Prometheus system user]
changed: [centos-node]
```

```
TASK [prometheus_centos : create Prometheus group] *****
ok: [centos-node]
```

Task 3: Create Prometheus Directories

```
- name: create /etc/prometheus
  ansible.builtin.file:
    path: /etc/prometheus
    state: directory
    mode: '0755'
    owner: prometheus
    group: prometheus

- name: create /var/lib/prometheus
  ansible.builtin.file:
    path: /var/lib/prometheus
    state: directory
    mode: '0755'
    owner: prometheus
    group: prometheus
```

```
TASK [prometheus_ubuntu : create /etc/prometheus] *****
ok: [ubuntu-node]
```

```
TASK [prometheus_ubuntu : create /var/lib/prometheus] *****
ok: [ubuntu-node]
```

```
TASK [prometheus_centos : create /etc/prometheus] *****
changed: [centos-node]
```

```
TASK [prometheus_centos : create /var/lib/prometheus] *****
changed: [centos-node]
```

Task 4: Download Prometheus

```
- name: download and extract Prometheus
  ansible.builtin.get_url:
    src: https://github.com/prometheus/prometheus/releases/download/v3.14.0/prometheus-3.14.0.linux-amd64.tar.gz
    dest: /tmp
    remote_src: true
    creates: /tmp/prometheus-3.14.0.linux-amd64.tar.gz
```

```
TASK [prometheus_ubuntu : download and extract Prometheus] *****
ok: [ubuntu-node]
```

```
TASK [prometheus_centos : download and extract Prometheus] *****
changed: [centos-node]
```

Verification

```
mdmlot@server1:~$ cd /tmp
mdmlot@server1:/tmp$ ls
prometheus-3.14.0.linux-amd64
snap-private-tmp
systemd-private-f48456d8063e4b9c99bb6058bfc4a2ab-apache2.serv
systemd-private-f48456d8063e4b9c99bb6058bfc4a2ab-color.d.serv
systemd-private-f48456d8063e4b9c99bb6058bfc4a2ab-fwupd.serv
systemd-private-f48456d8063e4b9c99bb6058bfc4a2ab-ModemManag
systemd-private-f48456d8063e4b9c99bb6058bfc4a2ab-power-profi
zYvkVK
systemd-private-f48456d8063e4b9c99bb6058bfc4a2ab-switcheroo
uGP
systemd-private-f48456d8063e4b9c99bb6058bfc4a2ab-systemd-log
systemd-private-f48456d8063e4b9c99bb6058bfc4a2ab-systemd-oo
systemd-private-f48456d8063e4b9c99bb6058bfc4a2ab-systemd-re
z
systemd-private-f48456d8063e4b9c99bb6058bfc4a2ab-systemd-ti
oT
systemd-private-f48456d8063e4b9c99bb6058bfc4a2ab-upower.serv
mdmlot@server1:/tmp$ cd prometheus*
mdmlot@server1:/tmp/prometheus-3.14.0.linux-amd64$ ls
LICENSE NOTICE prometheus prometheus.yml promtool
mdmlot@server1:/tmp/prometheus-3.14.0.linux-amd64$
```

Task 5: Install the Prometheus Binaries

```
- name: copy prometheus binaries to proper locations

  ansible.builtin.copy:
    src: "{{ item.src }}"
    dest: "{{ item.dest }}"
    remote_src: true
    owner: prometheus
    group: prometheus
    mode: '0755'
  loop:
    - { src: '/tmp/prometheus-3.14.0.linux-amd64/prometheus', dest: '/usr/local/bin/prometheus' }
    - { src: '/tmp/prometheus-3.14.0.linux-amd64/promtool', dest: '/usr/local/bin/promtool' }
```

```
TASK [prometheus_ubuntu : copy Prometheus binaries to /usr/local/bin/] *****
ok: [ubuntu-node] => (item={'src': '/tmp/prometheus-3.14.0.linux-amd64/prometheus', 'dest': '/usr/local/bin/prometheus'})
ok: [ubuntu-node] => (item={'src': '/tmp/prometheus-3.14.0.linux-amd64/promtool', 'dest': '/usr/local/bin/promtool'})
```

```
TASK [prometheus_centos : copy Prometheus binaries to /usr/local/bin/] *****
changed: [centos-node] => (item={'src': '/tmp/prometheus-3.14.0.linux-amd64/prometheus', 'dest': '/usr/local/bin/prometheus'})
changed: [centos-node] => (item={'src': '/tmp/prometheus-3.14.0.linux-amd64/promtool', 'dest': '/usr/local/bin/promtool'})
```

Verification

```
systemd [prometheus_ubuntu : copy Prometheus binaries to /usr/local/bin/] *****
mdm1ot@server1:/tmp$ cd /usr/local/bin
mdm1ot@server1:/usr/local/bin$ ls
mdm1ot@server1:/usr/local/bin$ ls
mdm1ot@server1:/usr/local/bin$ ls
mdm1ot@server1:/usr/local/bin$ ls
mdm1ot@server1:/usr/local/bin$ ls
prometheus promtool
mdm1ot@server1:/usr/local/bin$ nano prometheus
mdm1ot@server1:/usr/local/bin$ ls
prometheus promtool
mdm1ot@server1:/usr/local/bin$ prometheus
bash: /usr/local/bin/prometheus: Permission denied
mdm1ot@server1:/usr/local/bin$ ls
prometheus promtool
```

Task 6: Install the Prometheus Configuration Files

```
- name: copy Prometheus yaml file to /etc/prometheus/
  ansible.builtin.copy:
    src: '/tmp/prometheus-3.14.0.linux-amd64/prometheus.yml'
    dest: '/etc/prometheus/prometheus.yml'
    remote_src: true
    owner: prometheus
    group: prometheus
```

```
TASK [prometheus_ubuntu : copy Prometheus yaml file to /etc/prometheus/] *****
ok: [ubuntu-node]
```

```
TASK [prometheus_centos : copy Prometheus yaml file to /etc/prometheus/] *****
changed: [centos-node]
```

Verification

```
prometheus promtool
mdm1ot@server1:/usr/local/bin$ cd /etc/prometheus
mdm1ot@server1:/etc/prometheus$ ls
mdm1ot@server1:/etc/prometheus$ ls
mdm1ot@server1:/etc/prometheus$ ls
prometheus.yml
mdm1ot@server1:/etc/prometheus$
```

Task 7: Validate the Configuration

```
- name: Prometheus config file verification
  ansible.builtin.command: /usr/local/bin/promtool check config /etc/prometheus/prometheus.yml
  register: promtool_check
  changed_when: false
  failed_when: promtool_check.rc != 0
```

```
TASK [prometheus_ubuntu : Prometheus config file verification] *****
ok: [ubuntu-node]
```

```
TASK [prometheus_centos : Prometheus config file verification] *****
ok: [centos-node]
```

Task 8: Create the systemd Service

```
- name: create Prometheus systemd service file
  ansible.builtin.copy:
    src: ~/CPE212_loterte/midterm_skills/prometheus.service
    dest: /etc/systemd/system/prometheus.service
    owner: prometheus
    group: prometheus
```

```
TASK [prometheus_ubuntu : create Prometheus systemd service file] *****
ok: [ubuntu-node]
```

```
TASK [prometheus_centos : create Prometheus systemd service file] ***
changed: [centos-node]
```

Verification

```
ndm1ot@server1:/etc/prometheus$ cat /etc/systemd/system/prometheus.service
[Unit]
Description=Prometheus Time Series Collection and Processing System
Wants=network-online.target
After=network-online.target

[Service]
User=prometheus
Group=prometheus
Type=simple
ExecStart=/usr/local/bin/prometheus \
    --config.file=/etc/prometheus/prometheus.yml \
    --storage.tsdb.path=/var/lib/prometheus \
    --web.console.templates=/etc/prometheus/consoles \
    --web.console.libraries=/etc/prometheus/console_libraries

Restart=always

[Install]
WantedBy=multi-user.target
ndm1ot@server1:/etc/prometheus$
```

Task 9: Start and Enable Prometheus

```
- name: Start and enable Prometheus service
  ansible.builtin.systemd:
    name: prometheus
    state: started
    enabled: true
    daemon_reload: true
    become: true
```

```
TASK [prometheus_ubuntu : Start and enable Prometheus service] *****
changed: [ubuntu-node]
```

```
TASK [prometheus_centos : Start and enable Prometheus service] *****
changed: [centos-node]
```

Verification

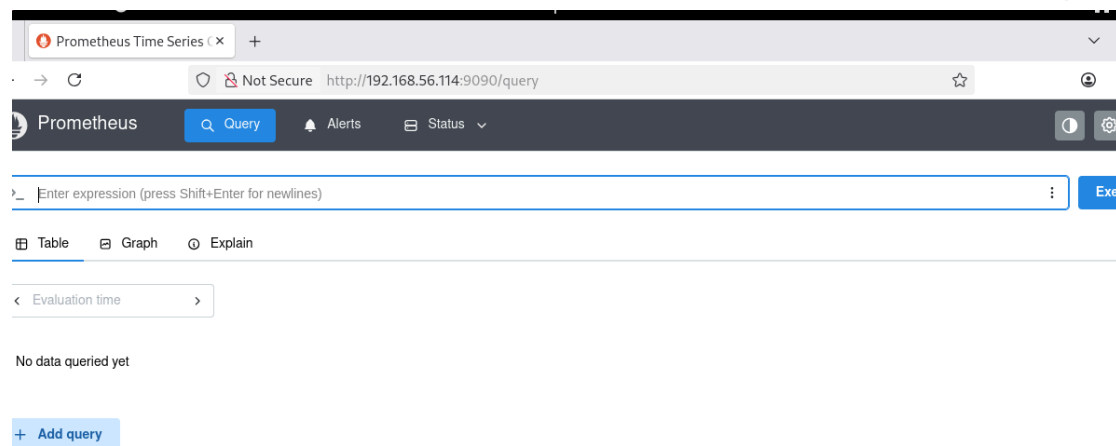
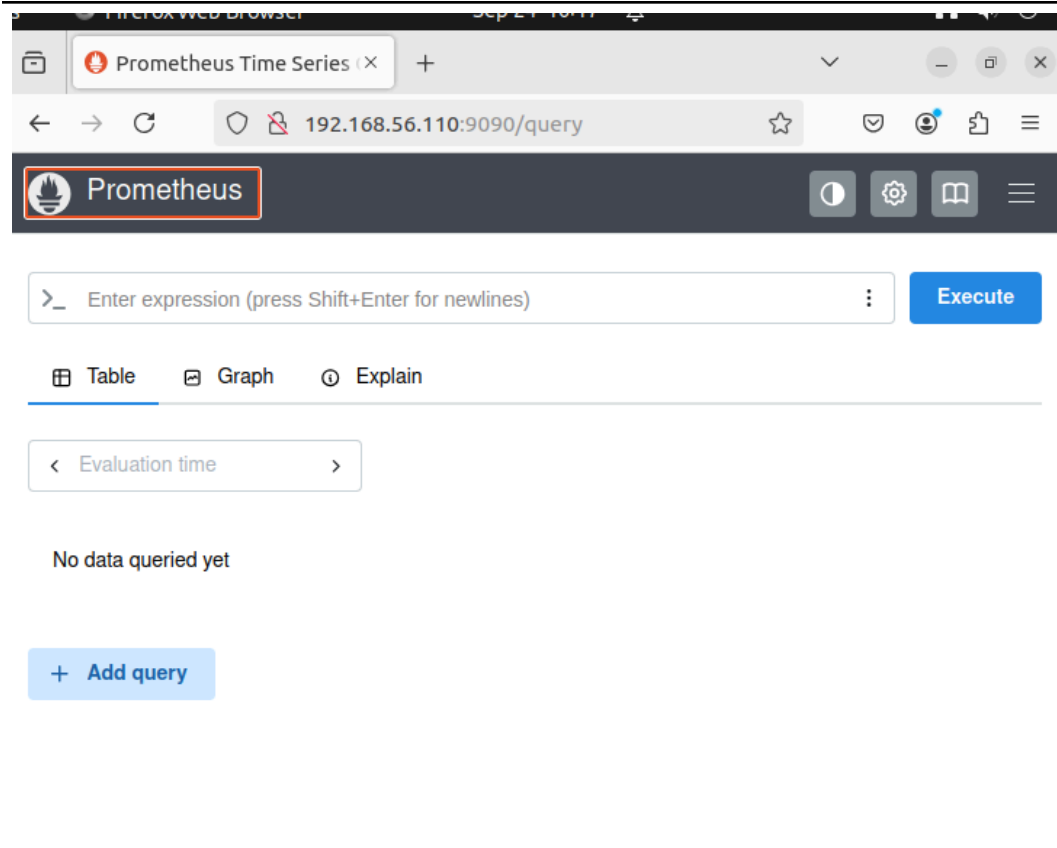
```
WantedBy=multi-user.target
mdm1ot@server1:/etc/prometheus$ systemctl status prometheus
● prometheus.service - Prometheus Time Series Collection and Processing System
   Loaded: loaded (/etc/systemd/system/prometheus.service; enabled; vendor p
   Active: active (running) since Thu 2026-09-24 10:10:13 PST; 3min 44s ago
   Main PID: 46844 (prometheus)
      Tasks: 7 (limit: 2260)
     Memory: 42.8M
        CPU: 1.546s
    CGroup: /system.slice/prometheus.service
           └─46844 /usr/local/bin/prometheus --config.file=/etc/prometheus/p>

Sep 24 10:10:14 server1 prometheus[46844]: time=2026-09-24T10:10:14.222+08:00 >
Sep 24 10:10:14 server1 prometheus[46844]: time=2026-09-24T10:10:14.228+08:00 >
Sep 24 10:10:14 server1 prometheus[46844]: time=2026-09-24T10:10:14.230+08:00 >
Sep 24 10:10:14 server1 prometheus[46844]: time=2026-09-24T10:10:14.246+08:00 >
Sep 24 10:10:14 server1 prometheus[46844]: time=2026-09-24T10:10:14.248+08:00 >
Sep 24 10:10:14 server1 prometheus[46844]: time=2026-09-24T10:10:14.249+08:00 >
Sep 24 10:10:14 server1 prometheus[46844]: time=2026-09-24T10:10:14.258+08:00 >
Sep 24 10:10:14 server1 prometheus[46844]: time=2026-09-24T10:10:14.296+08:00 >
Sep 24 10:10:14 server1 prometheus[46844]: time=2026-09-24T10:10:14.296+08:00 >
Sep 24 10:10:14 server1 prometheus[46844]: time=2026-09-24T10:10:14.298+08:00 >
lines 1-20/20 (END)
```

```
[mdm1ot@server3 ~]$ systemctl status prometheus
● prometheus.service - Prometheus Time Series Collection and Processing System
   Loaded: loaded (/etc/systemd/system/prometheus.service; enabled; preset: d
   Active: active (running) since Thu 2026-09-24 10:23:17 PST; 1min 54s ago
   Main PID: 7667 (prometheus)
      Tasks: 10 (limit: 22964)
     Memory: 43.8M (peak: 45.2M)
        CPU: 1.397s
    CGroup: /system.slice/prometheus.service
           └─7667 /usr/local/bin/prometheus --config.file=/etc/prometheus/pro>

Sep 24 10:23:18 server3 prometheus[7667]: time=2026-09-24T10:23:18.309+08:00 le>
Sep 24 10:23:18 server3 prometheus[7667]: time=2026-09-24T10:23:18.309+08:00 le>
Sep 24 10:23:18 server3 prometheus[7667]: time=2026-09-24T10:23:18.309+08:00 le>
Sep 24 10:23:18 server3 prometheus[7667]: time=2026-09-24T10:23:18.311+08:00 le>
Sep 24 10:23:18 server3 prometheus[7667]: time=2026-09-24T10:23:18.314+08:00 le>
Sep 24 10:23:18 server3 prometheus[7667]: time=2026-09-24T10:23:18.317+08:00 le>
Sep 24 10:23:18 server3 prometheus[7667]: time=2026-09-24T10:23:18.317+08:00 le>
Sep 24 10:23:18 server3 prometheus[7667]: time=2026-09-24T10:23:18.380+08:00 le>
Sep 24 10:23:18 server3 prometheus[7667]: time=2026-09-24T10:23:18.380+08:00 le>
Sep 24 10:23:18 server3 prometheus[7667]: time=2026-09-24T10:23:18.381+08:00 le>
lines 1-20/20 (END)
```

Task 10: Verify Prometheus and Access Prometheus Through a Browser



2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: _____

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: _____

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: _____

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